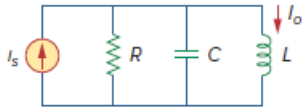


Problem

Show that the parallel RLC circuit shown in Fig. 16.105 is stable.

Figure 16.105:



Step-by-step solution

Step 1 of 2

If V_o is the voltage across R , applying KCL at the non reference node gives

$$I_s = \frac{V_o}{R} + SCV_o + \frac{V_o}{SL} = \left(\frac{1}{R} + SC + \frac{1}{SL} \right) V_o$$

$$V_o = \frac{I_s}{\frac{1}{R} + SC + \frac{1}{SL}} = \frac{SRLI_s}{SL + R + S^2RLC}$$

$$I_o = \frac{V_o}{R} = \frac{SLI_s}{S^2RLC + SL + R}$$

Step 2 of 2

$$H(S) = \frac{I_o}{I_s} = \frac{SL}{S^2RLC + SL + R} = \frac{\frac{S}{RC}}{\frac{S^2 + S}{RC} + \frac{1}{LC}}$$

The Roots

$$S_{1,2} = \frac{-1}{2RC} \pm \sqrt{\frac{1}{(2RC)^2} - \frac{1}{LC}}$$

Both lie in the left half plane since R, L, C are positive quantities.

Thus, circuit is stable